

Practice Quiz: Collective Intelligence — Module 02:

Agents

Name: ____ Date: ____

Part A: Multiple Choice

1. Collective agency emerges when: A) A manager assigns tasks B) Individual agents align models, distribute inference, and develop coordination producing behavior no individual could achieve alone C) The team has a large budget D) More people are hired
2. **Distributed cognition** (Hutchins) means: A) Everyone thinks the same thing B) The cognitive task is distributed across people, tools, and environmental artifacts C) Knowledge is stored in the cloud D) Teams use distributed computing
3. Woolley et al. found that collective intelligence depends primarily on: A) The team's average IQ B) Social sensitivity, equal participation, and cognitive diversity C) The leader's charisma D) Budget size
4. **Social loafing** occurs because: A) People are lazy B) Individual agents reduce active inference when they believe others will compensate C) Teams are too small D) Work is too interesting
5. **Knowledge hiding** undermines collective intelligence by: A) Making work easier B) Treating shared information as a competitive resource, depriving the collective model of critical inputs C) Reducing communication overhead D) Protecting intellectual property
6. A team's collective intelligence is best measured by: A) Sum of individual IQ scores B) The team's ability to solve problems no individual member could solve alone C) Team satisfaction surveys D) The number of meetings held

7. **Status hierarchy effects** distort collective agency by: A) Motivating high-performers
B) Weighting evidence by speaker status rather than evidence quality C) Creating clear decision-making D) Reducing conflict
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Part B: Short Analysis

1. A team of 8 highly talented engineers consistently produces mediocre results while a team of 8 average engineers in the same company delivers exceptional work. Diagnose this paradox using the concepts of collective agency. What specific collective intelligence factors likely differ between the teams?
2. Your team needs to make a critical architectural decision. The most senior engineer has strong views, and junior members tend to defer. Design a decision process that harvests the collective intelligence of the entire team while managing status hierarchy effects.
3. You're building a new cross-functional team to tackle an ambiguous strategic challenge. Using the conditions for collective intelligence (Woolley), specify: team composition (roles/backgrounds), team norms, meeting design, and communication protocols.